Turns of a T-Mesh: Obstructions to Closing Edge Loops

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Marc Gardent*
Independent researcher
ORCID: 0009-0000-6466-3098
Contact: through the repository, `https://github.com/marcgardent/openclare`

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Abstract

In the quantization of a T-mesh, an edge loop closes into a ring when a *turn*, a straight walk across the patches that returns to its first side, has zero *shift*, an integer linear form in the arc lengths; otherwise it winds as a helix. We study why turns cannot all be closed. Reading the shift as translational holonomy of the flat metric, we show that for a simple turn it is the across component of the Burgers vector of the singular vertices on one side; that a 3–5 pair or an exchange of two vertices, joined by a staircase of arcs, forces one of two turns to be shifted, and a separatrix that winds once round a single turn; and that on a closed sphere the shift and the length of a turn have the same parity. Fox calculus extends the Burgers sum to any combination of turns, and *combs*, local dislocations carried by single arcs, write every certificate of a conflict found on a human body. The conflicts bound from below the number of turns that every quantization within given bounds leaves shifted: on that body, within the bounds of separation of a remeshing engine, exactly one to nine of the turns it holds, where it leaves 13 to 123 shifted. Choosing the turns to close is NP-hard, already for the bidirected balances and 0–1 bounds of quantization. The algebraic cores of the proofs are checked in Lean.

1 Introduction

Automatic quad remeshing often produces edge loops that wind round a part of the shape as helices instead of closing into rings [4]. Whether a loop closes is decided when a quantizer assigns whole lengths to the arcs of a T-mesh traced in a seamless parametrization [6, 17, 29]: a loop follows a *turn*, a straight walk across the patches that returns to its first side, and closes when the *shift* of the turn, an integer linear form in the arc lengths, vanishes. A companion paper [14] adds these equations to a quantizer; on real models they cannot all hold together with the bounds that keep singular vertices apart. This paper asks why, and what follows.

Sections 3 to 7 go from the holonomy of one turn to conflicts between turns, any combination of turns, and borders and handles; Sections 8 and 9 bound the turns every quantization leaves shifted and show that choosing them is NP-hard; Sections 10 and 11 say what is machine-checked and what is open. The ingredients are classical (holonomy of flat cone metrics, Burgers vectors, the parity of quadrangulations, Fox calculus, covers of infeasible subsystems); their statements

*This note is distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). The code of openclare and the Lean proofs are under the GPL-3.0-or-later of the repository. The note describes the Lean proofs at commit `6848ef55c9ba` of `https://github.com/marcgardent/openclare`. A French translation is in `paper.fr.pdf`, beside this note.

for turns, the lower bound on helices and the hardness for the instances of quantization are ours, as far as we found. Statements are *proved* when they come with a proof, *measured* when they come from exact arithmetic or solver output on the systems the remesher openclare exports, and *conjectured* otherwise.

This work began as a programming project, openclare, an open-source quad remesher, and became research when engineering alone could no longer tell which results could be guaranteed and which computations saved. The results here answer from the side of what cannot be done: a repair that keeps the length of a turn of odd shift cannot close it on a closed sphere; a turn whose shift is a sign-definite form with an arc held open stays a helix and need not be tried; the turns every quantization leaves shifted measure what a heuristic can still gain; and the hardness of choosing them is why an engine settles for a heuristic.

2 Related work

The valences of a closed quad mesh of genus g satisfy $\sum_v (4 - k(v)) = 8(1 - g)$, which says how many singular vertices are needed, not where. A closed quad mesh induces a quartic differential whose zeros and poles are its singular vertices; Lei et al. [27] list as open the guarantee that all its trajectories are finite, and a trajectory that is not finite is an edge loop that does not close. A quadratic differential whose trajectories all close is Jenkins–Strebel [31], a union of flat cylinders; a helix is a cylinder glued back with a shift of whole rows. Chen et al. [7] characterize the flat metrics of quad meshes, assuming that the lines of the field are finite loops. Izmestiev et al. [19] prove non-existence results for quad meshes, among them the torus with one vertex of valence three and one of valence five [3], with the holonomy of the developing map, whose translation, when the rotation is trivial, is the Burgers vector of crystallography; a pair of opposite disclinations is a dislocation [26]. The periods of a flat cone metric are local coordinates of it [33, 34]. Bommes et al. [4] remove helices from finished meshes; stripe parameterizations of tubes [22] round the gap round a tube, our shift, to a whole number. We found no lower bound on the number of helices for a given singularity configuration; ours is a minimum cover of irreducible infeasible subsystems [2, 30].

3 Turns, shifts and holonomy

A T-mesh is a partition of the surface into rectangles, the *patches*, whose sides are made of *arcs*; nodes where a trace stops on another are T-junctions, and the singular vertices of the field are nodes. Each patch P has local coordinates with its first corner at the origin and its sides along the axes. A quantization assigns every arc a a length $\ell_a \in \mathbb{Z}_{\geq 0}$ such that opposite sides of every patch have equal length, the *balances* $B\ell = 0$; the quad mesh is then the integer grid of each patch.

Definition 1 (Turn)**.** *A* turn γ *is a straight walk across patches, from an offset of its first side that meets no node: it enters a patch through a side, leaves through the opposite side at the same distance from the corresponding corner, enters the neighbouring patch across the arc it reaches, and so on, until it comes back into its first patch through its first side. The rotation of the composed crossings is then the identity, since the walk re-enters its first side inward.*

Crossing from patch P to patch Q across an arc composes a quarter-turn rotation of the local frame with a translation whose components are signed sums of arc lengths along the rims of P and Q.

Proposition 1 (The shift of a turn is linear)**.** *For a turn* γ *there are integer coefficients* c_a*, independent of the lengths, such that after a full turn a point of the first side has moved across*

the walking direction by

$$s_\gamma(\ell) = \sum_a c_a\, \ell_a.$$

For integer lengths ℓ, let o be a half-integer offset of the first side whose walk at the lengths ℓ crosses the same patches as γ (we call the set of such offsets the corridor *of γ at ℓ). Then the dual line of the quad mesh at o comes back at $o + s_\gamma(\ell)$: it closes after one round if and only if $s_\gamma(\ell) = 0$. If moreover $o + j\, s_\gamma(\ell)$ lies in the corridor for $j = 1, \dots, t-1$, it winds t times round as a helix of pitch $|s_\gamma(\ell)|$ in the sense of [4]. If the corridor holds no half-integer, no strip of the mesh follows γ.*

Sketch. Each crossing is a map $x \mapsto R_k x + t$ with R_k a rotation by k quarter-turns and t a vector whose entries are sums of $\pm\ell_a$. The composition along γ has trivial rotation, hence is a translation $T_\gamma(\ell)$, linear in ℓ with integer coefficients; its component across the walking direction is $s_\gamma(\ell)$. Singularities are nodes, so the quad mesh is a regular grid along the walk, and a dual line at distance o from the first corner that crosses the same patches comes back at $o + s_\gamma(\ell)$. The corridor, two inequalities per crossing, may contain o but not $o + s_\gamma(\ell)$; the next round then follows other patches. □

The component of $T_\gamma(\ell)$ along the walk is the *length* $\alpha_\gamma(\ell)$ of the turn, which at whole lengths is the number of quads of the strip in one turn. At the real lengths of the parametrization the shift is the *real shift*, and its share of the length the *lean*. At whole balanced lengths with positive patch sides every strip of the quad mesh is such a walk [14, Prop. 2], and every shift is whole; closing a ring asks it to be zero.

Flat metric and holonomy. Unless stated otherwise the surface Σ is closed, and *genus zero* means a closed sphere. Balanced lengths $\ell > 0$ make each patch a Euclidean rectangle, and Σ a flat surface with cone points at the singular vertices: a vertex of valence k has angle $k\pi/2$ and curvature $(4-k)\pi/2$, and T-junctions are not cone points. Let Σ° be Σ without its singular vertices, x_0 a point inside a patch, and R the positive quarter-turn. Developing a loop γ of Σ° based at x_0 patch after patch gives its *holonomy*, a motion $\hat{h}(\gamma)\colon x \mapsto R^j x + t$ of the coordinates of the first patch; it depends only on the homotopy class of γ, and $\hat{h}(\alpha\beta) = \hat{h}(\alpha) \circ \hat{h}(\beta)$ [19]. A small loop round a singular vertex c of valence k, run with c on its right and reached along a path that develops it at p, has holonomy $x \mapsto R^{4-k}(x-p) + p$. When the rotation of $\hat{h}(\gamma)$ is the identity, its translation is the *Burgers vector* of γ. For a turn γ, let $\bar{\gamma}$ be the closed curve that follows the walk and, back in its first patch, returns to its start along a segment parallel to the first side; with e_1 the walking direction and e_2 the direction along the first side, $\hat{h}(\bar{\gamma})$ is a translation. It continues the chart along the loop, while the crossings of Proposition 1 develop the last patch into the first: $\hat{h}(\bar{\gamma})$ is the inverse of $T_\gamma(\ell)$. From here on we read shifts in the convention of the holonomy, $s_\gamma(\ell) = \langle e_2, \hat{h}(\bar{\gamma}) \rangle$, the opposite of the shift of Proposition 1; changing the sign of every shift changes no statement about rings, parities or the existence of certificates, and the signs of measured identities are those of the exported shifts. Shifts, developed positions and Burgers vectors are integer linear forms in ℓ as long as the paths cross the same patches, so an identity between them that holds on an open set of balanced positive lengths holds for all balanced lengths; for whole lengths the nodes develop onto $\mathbb{Z}^2$. A form f equals g *modulo the balances* if $f - g$ is a combination of the rows of B.

Data. The measured statements refer to the T-mesh of the male body of Blender's Essentials library remeshed by openclare at 10,590 quads, exported with every turn (branch `rd-exact`, engine before commit `3b3122d`) and studied in the companion paper [14, §7]: 261 singular vertices (133 of valence three, 127 of valence five, one of valence two) among 1,295 nodes, 2,069 arcs, 776 patches, 1,552 balances. Turns are found in eight *ways*, way $m + 2j + 4i$ walking from the shares $\{0.5\}$ ($m = 0$) or $\{0.2, 0.5, 0.8\}$ ($m = 1$) of each side and *holding* (trying to close)

a turn if its real shift is above 0.05, its lean below 0.05 ($j = 0$) or 0.12 ($j = 1$) and its length below 2 ($i = 0$) or 6 ($i = 1$) sides of the T-mesh; the engine keeps way 2. The ways find 1,299 turns and hold 765 (summed over the ways). The integer search holds 28 to 36 arcs at length at least one, the *bounds of separation*; P_{eng}, the balanced lengths meeting them, is neither a subset nor a superset of the valid quantizations, which only ask a chain between two singular vertices to keep a positive length. Identities are checked in floating point unless said exact; programs were solved with HiGHS 1.8.0 [18] through SciPy 1.16.3 [35], on an Intel Core i7-7700K (4 cores, 2017, 62 GB) shared with other jobs.

4 Turns and singularities

Lemma 1 (Gauss–Bonnet for turns)**.** *Let γ be a turn whose curve $\bar{\gamma}$ is simple. If $\bar{\gamma}$ bounds a disc $D \subset \Sigma$, the singular vertices inside D satisfy $\sum_{c \in D}(4 - k(c)) = 4$. If the curves of two turns are disjoint and bound an annulus A, then $\sum_{c \in A}(4 - k(c)) = 0$.*

Proof. For a region of a flat surface with cone points bounded by straight segments, Gauss–Bonnet reads $\sum_c \kappa_c + \sum_i \theta_i = 2\pi\chi$, with $\kappa_c = (4 - k(c))\pi/2$ the curvatures inside and θ_i the turning angles of the border [5]. The curve $\bar{\gamma}$ turns only at the ends of its closing segment, by $+\pi/2$ and $-\pi/2$, so the turning angles sum to zero on each curve, seen from either side; $\chi(D) = 1$ and $\chi(A) = 0$. □

On a closed sphere a simple turn thus splits the singular vertices into two groups of total index four, each of at least two vertices: a fingertip beyond a ring holds the equivalent of four vertices of valence three.

Lemma 2 (One self-crossing)**.** *Let the surface be closed of genus zero, and let the curve $\bar{\gamma}$ of a turn cross itself exactly once, transversally, with no singular vertex on it. It cuts the surface into three discs: two* lobes*, each bounded by one of the two loops of $\bar{\gamma}$, and a third region bounded by both. If the crossing lies on the walk, the totals of $4 - k$ over the singular vertices of the lobes and the third region are* 3*,* 3 *and* 2*; if it lies on the closing segment, they are* 2*,* 2 *and* 4*.*

Proof. The walk crosses itself at a right angle, so each loop leaves the crossing X along one passage and comes back along the other, its two rays a quarter-turn apart. Each loop is simple and bounds a disc containing the quadrant between its rays, a lobe with a corner of angle $\pi/2$ at X; the lobes take opposite quadrants and the third region the other two, and the three faces of this one-vertex graph on the sphere are discs. By Gauss–Bonnet a disc whose border turns by θ_i holds total index $4 - \sum_i \theta_i/(\pi/2)$, which gives 3, 3 and 2 before the corners of the closing segment, of turnings $+\pi/2$ and $-\pi/2$ in succession. If the crossing lies on the walk, both are on one loop, every region sees both, and they cancel. Otherwise let the walk start at A along e_1, come back at Z and cross the closing segment between them; it crosses along $-e_1$, since along e_1 the turn would have ended there. The loop through A then turns the same way at A as at X, and so does the loop through Z at Z and at X (a check of the four directions): each lobe sees a second convex corner and holds 2, and the third region the rest of the total 8. □

Proposition 2 (The shift is set by the singularities)**.** *Let γ, γ_1, γ_2 be turns with simple curves.*

(a) Let $\bar{\gamma}$ bound a disc D holding the singular vertices $c_1, \dots, c_m$, and let x_0 be a point of $\bar{\gamma}$, the start of the walk for instance. Let $T \subset D$ be a tree whose vertices include $c_1, \dots, c_m$ and which meets $\bar{\gamma}$ only at x_0. Cut along T, the disc D becomes a disc Ω without cone point, which develops into the plane, with the frame (e_1, e_2) at x_0. Number the c_i in the order in which a walk round T from x_0, keeping T on its right, first reaches them, and let p_i be the developed position of c_i at that first visit. Then $s_\gamma(\ell) = \pm\langle e_2, B_D \rangle$, with

$$B_D = \sum_{i=1}^{m} Q_i \, (I - R_i) \, p_i, \qquad R_i = R^{4-k(c_i)}, \qquad Q_i = R_m R_{m-1} \cdots R_{i+1},$$

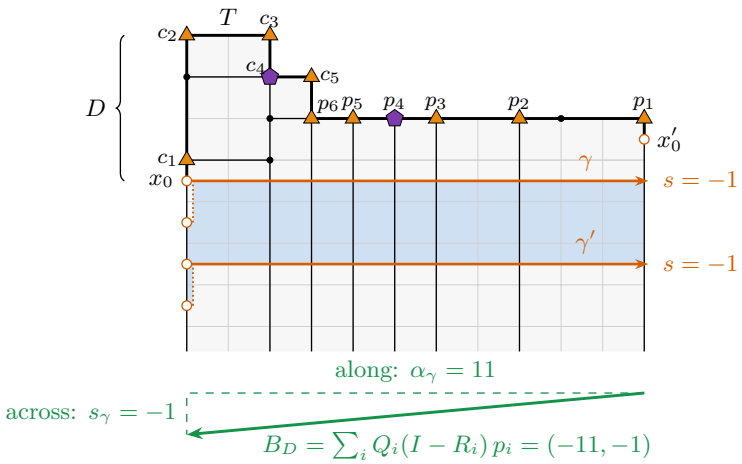


Figure 1: The shift read on the singularities (Proposition 2). A cap D with six singular vertices (valences $3, 3, 3, 5, 3, 3$, total index 4), cut along a tree T and developed; each c_i is drawn at its first visit p_i on the walk round T. The turn γ, of length 11, comes back one row lower, and $B_D = \sum_i Q_i(I - R_i)p_i = (-11, -1)$: its along component is the length, its across component the shift. The turn γ' bounds with γ an annulus without singular vertex (shaded) and has the same shift (part (c)); $11 \equiv -1 \pmod 2$ (Proposition 5). In all figures the shifts are labelled in the convention of Proposition 1, offsets measured upwards.

the translation of $\rho_m \circ \cdots \circ \rho_1$, where $\rho_i(x) = R_i(x - p_i) + p_i$; the sign is $+$ if D lies on the right of $\bar\gamma$, and $-$ otherwise.

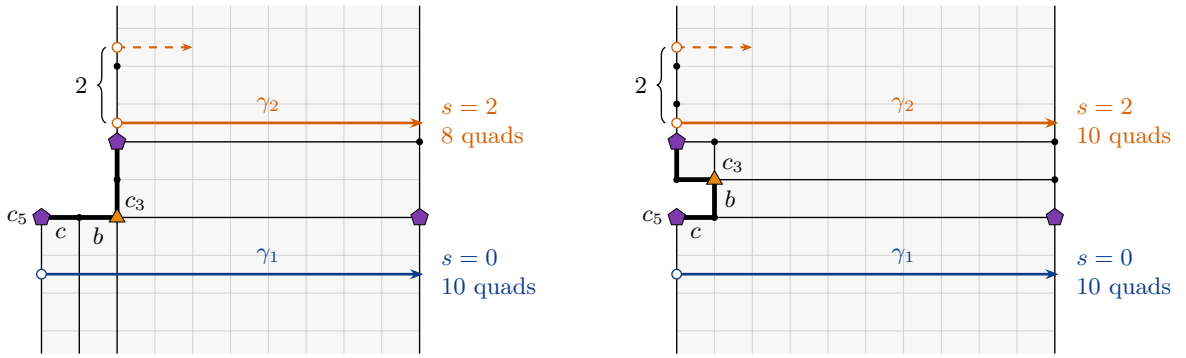
(b) Let $\bar\gamma_1$ and $\bar\gamma_2$ be disjoint and bound an annulus A, round which the two turns go the same way, and let a path κ inside A from $\bar\gamma_1$ to $\bar\gamma_2$ carry the walking direction of γ_2 onto that of γ_1 (the turns are parallel*); measure both shifts along the direction e_2 carried by κ. Then $s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \langle e_2, B_A\rangle$, where B_A is the Burgers vector of the border of the disc A cut along κ, given by the same sum over the singular vertices inside A, developed inside the cut disc.*

(c) In particular, parallel turns bounding an annulus without singular vertex have the same shift for every ℓ: a helix is a property of a whole cylinder of parallel turns, not of one of them.

The order matters: on Figure 1, walking round T the other way gives $(1, 1)$ in place of $B_D = (-11, -1)$.

Proof. (a) Orient each edge f of T away from x_0; its two sides on the border of Ω are glued by a motion G_f, from the side the walk round T follows first to the other. Let f end at v, from which $f_1, \dots, f_r$ leave in the order of the walk. A small arc turning clockwise round v from the first side of f to the second crosses $f_1, \dots, f_r$ and turns through the whole angle at v, so $G_f = G_{f_r} \circ \cdots \circ G_{f_1} \circ \rho_v$, with $\rho_v(x) = R^{4-k(v)}(x - p_v) + p_v$, p_v the position at the first visit ($\rho_v = I$ if v is not singular). At x_0, with $\bar\gamma$ run with D on its right in place of the arc, $\hat h(\bar\gamma) = G_{f_r} \circ \cdots \circ G_{f_1}$. Unfolding the recursion composes the ρ_{c_i} in the order of first visit; the rotation is $R^{\sum_i (4-k(c_i))} = I$ by Lemma 1, expanding gives B_D, and with D on its left $\bar\gamma$ has the inverse holonomy. (b) With x_0 on $\bar\gamma_1$ at the start of κ, the loop $\lambda = \bar\gamma_1\, \kappa\, \bar\gamma_2^{-1} \kappa^{-1}$ borders the disc A cut along κ. Developing along κ reads $\hat h(\bar\gamma_2)$ in the coordinates at x_0, where the walk of γ_2 runs along e_1; so $\hat h(\lambda)$ is the translation by $\hat h(\bar\gamma_1) - \hat h(\bar\gamma_2)$, and part (a) applies to the cut disc, whose singular vertices have total index zero. (c) The sum is empty. □

In the T-meshes of openclare, whose traces start at singular vertices, an annulus without singular vertex holds no node unless a trace winding round enters it through a closing segment,



(a) straight chain (b) staircase

Figure 2: A 3–5 dislocation between a ring γ_1 and a helix γ_2 of a band, cut open along a trace of the vertex of valence five, which is drawn three times. (a) The pair is joined along one trace by two arcs, $\delta = \ell_b + \ell_c = 2$: γ_2 comes back 2 rows higher, with 8 quads per turn, and the Burgers vector of the pair is $(2, 2)$ (Proposition 3). (b) The pair is joined by a staircase of two arcs, $d = \varepsilon = (-1, -1)$: γ_2 comes back 2 rows higher, with 10 quads (Proposition 6). In both, the shifts differ by $\ell_b + \ell_c = 2$, and both turns are rings only if the two vertices merge.

so two turns as in (c) usually lie in one corridor.

The simplest configuration between two parallel turns is a pair of valences three and five, which lets the number of edges round a limb change. A pair of opposite disclinations of rotation Ω is a dislocation with a Burgers vector of $2\sin(\Omega/2)$ times their separation [26, Eq. (5)]. We read it on the shifts of two turns (Figures 1 and 2).

Proposition 3 (A dislocation between two turns)**.** *In the setting of Proposition 2(b), let the annulus A hold exactly two singular vertices, c_3 of valence three and c_5 of valence five, and let $d = p_5 - p_3 = d_1 e_1 + d_2 e_2$ be their separation, developed inside the cut disc. Then:*

(i) $s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \varepsilon_1 d_1 + \varepsilon_2 d_2$ with signs $\varepsilon_1, \varepsilon_2 \in \{+1, -1\}$ that depend on the configuration but not on ℓ, and $|B_A| = \sqrt{2}\,|d|$. The two turns can close together only if c_5 lies, as seen from c_3, on one particular diagonal of the field.

(ii) If c_3 and c_5 lie on one straight line of the field inside A, of length δ (in a T-mesh: they are joined by a chain of arcs along one trace, and δ is the sum of their lengths), and some path κ as in Proposition 2(b) does not cross that line, then $|s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell)| = \delta$. The two turns can both be rings only if $\delta = 0$, that is if the two singularities merge; as long as the bounds of separation keep $\delta \geq 1$, one of them at least is a helix.

(iii) For whole lengths, $s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) \equiv d_1 + d_2 \pmod 2$: a pair at an odd number of edges from one another, counted along the two directions of the field, never lies between two rings.

Proof. With $R_3 = R$ and $R_5 = R^{-1}$, the two orders give $B_A = R^{-1}(I - R)p_3 + (I - R^{-1})p_5 = (I - R^{-1})d$ or $B_A = R(I - R^{-1})p_5 + (I - R)p_3 = (I - R)(-d)$. With $R(a, b) = (-b, a)$, $(I - R^{-1})(d_1, d_2) = (d_1 - d_2,\ d_1 + d_2)$ and $(I - R)(-d_1, -d_2) = (-d_1 - d_2,\ d_1 - d_2)$; reversing the orientation of the border inverts the rotations and gives the two other sign patterns (checked symbolically). In all four both components are $\pm d_1 \pm d_2$ and $|B_A|^2 = 2|d|^2$, which is (i). For (ii), take κ not crossing the segment joining c_3 to c_5 (crossing it would turn the frame by a quarter-turn round c_3) and a tree containing that segment, and develop c_5 from c_3 along it: d is δ times one of $\pm e_1$, $\pm e_2$. For (iii), $d \in \mathbb{Z}^2$ and $\pm d_1 \pm d_2 \equiv d_1 + d_2$. □

Proposition 4 (Checkerboard rule)**.** *In the setting of Proposition 2(b), with whole lengths, colour the points $p \in \mathbb{Z}^2$ of the cut disc black or white by the parity of $p_1 + p_2$ (either colour may be called white). Then $s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell)$ has the parity of the number of singular vertices of odd valence inside A that develop onto white points. In particular, two rings bound an annulus only if that number is even.*

Proof. In the sum of Proposition 2, a vertex of even valence has $R_i = \pm I$ and $(I - R_i)p_i \in \{0, 2p_i\}$; for odd valence both components of $(I - R^{\pm 1})(a, b) = (a \pm b,\ b \mp a)$ have the parity of $a + b$. Quarter-turns keep the property that both components have a given parity, and sums add the parities, which gives the count. A quarter-turn keeps colours; a whole translation with $t_1 + t_2$ odd swaps them, which keeps the parity of the count because the vertices of odd valence inside A are even in number, their odd indices summing to zero with the even ones (Lemma 1). Developing a vertex along another path inside the cut disc rotates it about another singular vertex, which keeps its colour. □

The rule also holds on 13,000 random configurations checked with SymPy.

Parity on a sphere. Each turn of a closed sphere carries its own parity condition, which needs neither a simple curve nor the holonomy.

Proposition 5 (Parity of a turn on a sphere)**.** *Let the surface be closed of genus zero and the T-mesh admit lengths that are positive and balance every patch (its real lengths do). For every turn γ and all whole lengths ℓ balancing every patch, of any sign,*

$$\alpha_\gamma(\ell) \equiv s_\gamma(\ell) \pmod 2.$$

Proof. First let ℓ be whole, balanced and positive: the patches form a quad mesh of the sphere, whose graph is bipartite, since $H_1(S^2; \mathbb{Z}/2) = 0$ makes every cycle a sum of faces of four edges. In a 2-colouring the colour of the point (i, j) of a patch P is $i + j + \kappa_P \bmod 2$. For $M(x) = R^k x + t$, $t \in \mathbb{Z}^2$, the coordinate sum of $M(x)$ has the parity of that of x plus $\pi(M) = t_1 + t_2$. Let $P_0, \dots, P_N = P_0$ be the patches crossed by γ and D_k the motion developing P_k into P_0, with $D_N = T_\gamma(\ell)$. A node of the arc from P_k to P_{k+1}, of coordinates y, y' with $D_k(y) = D_{k+1}(y')$, has the colour $(D_k y)_1 + (D_k y)_2 + \pi(D_k) + \kappa_{P_k}$, and likewise from P_{k+1}; so $\kappa_{P_k} + \pi(D_k)$ is constant, and at $k = N$, $\pi(T_\gamma(\ell)) \equiv 0$. In the frame of the first side, that of P_0 up to a quarter-turn and signs, $T_\gamma(\ell) = (\alpha_\gamma(\ell), \pm s_\gamma(\ell))$. For general whole balanced ℓ: the balanced lengths form a rational subspace with positive real points, so there are positive whole balanced lengths p, and the linear form $\alpha_\gamma + s_\gamma$ takes at $\ell = (\ell + Np) - Np$, N large, the difference of two even values. □

The proof uses only that the quad graph is bipartite; it fails in higher genus (a 3×3 grid on a torus has rings of three quads) and with borders (Theorem 1). On a closed sphere it gives at once:

Corollary 1. *On a closed surface of genus zero:*

(a) every ring of quads has an even number of quads;

(b) a helix of odd pitch winds round an odd number of quads per turn;

(c) closing a helix of odd pitch changes the length of its walk by an odd number of edges: a repair that keeps the length of a turn cannot close it if its pitch is odd, and changing the pitch by an odd amount always changes the length.

Proof. (a) A ring is the strip of a walk with $s = 0$ [14, Prop. 2], so α is even. (b) and (c) are the proposition read before and after the change. □

Part (a) is the classical bipartiteness of quadrangulations of the sphere [21, Lemma 3.1]. For a turn bounding a disc, the argument of the checkerboard rule gives both components of $B_D = (\alpha_\gamma, s_\gamma)$ the parity of the number of vertices of odd valence on white points, which is the proposition for that turn.

Remark 1 (Borders; a second proof)**.** *The proposition and part (a) of the corollary need the surface to be closed: an open cylinder of three quads round is a ring of three quads. A second proof, from the formalization, needs neither the bipartite graph nor positive lengths: every loop of Σ° is a product of loops round the singular vertices (Section 6), quarter-turns about whole points that keep the parity of $p_1 + p_2$, so the translation of every loop has an even coordinate sum.*

How many independent rings. Rings that do not cross are disjoint simple closed curves of Σ°; on a closed sphere with $n \geq 3$ singular vertices at most $n-3$ are pairwise non-homotopic, as for the cylinders of a Jenkins–Strebel differential [31], and homotopic ones share their shift, so their closing conditions have rank at most $n-3$ ($3g-3+n+b$ plus at most b classes of collars with genus g and b borders).

Certificates. Closing a set of turns within bounds is a question of linear feasibility.

Remark 2 (Sign-definite turns; Farkas certificates)**.** *If the shift of a turn equals, modulo the balances, a form $\sum_a c_a \ell_a$ with every $c_a \geq 0$, then for all balanced $\ell \geq 0$ the shift $s_\gamma(\ell)$ is never negative, and it is zero only if every arc with $c_a > 0$ has length zero. If one of those arcs is held open by a bound of separation, the turn is a helix under every quantization meeting the bounds; it should never be held, and one linear program per turn finds such turns. More generally, by Farkas' lemma (with $x = b + z$, $z \geq 0$), the system $B\ell = 0$, $s_\gamma(\ell) = 0$ for $\gamma \in H$, $\ell \geq b$ has no solution exactly when some combination $\sum_{\gamma \in H} u_\gamma s_\gamma$ equals, modulo the balances, a form $\sum_a c_a \ell_a$ with $c \geq 0$ and $\sum_a c_a b_a > 0$; and an arc a has length zero in every solution with $\ell \geq 0$ exactly when some such combination has $c_a > 0$. The data being integral, bounds included, the multipliers of the shifts and of the balances can be taken whole.*

Remark 3 (Whole solutions exist at some size)**.** *The balance and closing constraints are homogeneous and the bounds of separation are lower bounds $\ell \geq b$ with $b \geq 0$. If they have a real solution they have a rational one, a vertex of a rational polyhedron, and multiplying it by the common denominator N gives whole lengths $N\ell \geq Nb \geq b$ that satisfy them all. Closing in whole numbers is thus never impossible in itself; what can fail is closing* at a given size *(number of quads) or within a given deviation from the targets.*

On the male body. The statements of this section hold wherever they apply (measured). The crossing maps give the exported shift of all 1,299 turns. On the 1,436 sides of the 718 simple ones, of index 4 and 4 to 257 singular vertices, the holonomy equals the sum of Proposition 2(a) in both components, while the order of last visit fails on 150 of the 158 discs of way 2; the checkerboard rule holds there and on the 30,148 annuli between disjoint simple turns, of index 0; the 294 pairs with no singular vertex between them have equal shifts; on the ten annuli holding one 3–5 pair, Proposition 3(i) holds, the two never on one trace. Proposition 5 holds on all turns and on the 1,838 ways round of the strips of the eight meshes. Of the 101 turns crossing themselves once, 97 split as $(3, 3, 2)$ and 4, crossing their closing segment, as $(2, 2, 4)$; 480 cross themselves up to 107 times, and only 356 of the 765 held turns are simple: Section 6 handles any curve.

ways	size	turns	multipliers	bounded arcs	exact check
0, 1, 2, 4	1	42 (way 0)	-1	1937	way 0
1, 3, 5, 7	2	54, 61 (way 1)	$1, -1$	1587	—
1, 3	2	45, 49 (way 1)	$1, -1$	1069	—
1, 3, 7	2	26, 37 (way 1)	$-1, 1$	619	—
5, 7	2	58, 94 (way 5)	$1, -1$	858	—
5	2	49, 71	$-1, 1$	619, 625	—
6	2	57, 58	$1, 1$	1241	—
7	2	8, 15	$1, 1$	191	—
1	3	36, 37, 39	$1, 1, -1$	619	yes
1	4	40, 56, 59, 61	$1, 1, -1, -1$	1587	yes
1	4	66, 71, 75, 76	$-\frac{7}{6}, -\frac{5}{6}, -\frac{1}{3}, -\frac{1}{2}$	1967, 1980, 1998	yes
3	4	73, 78, 82, 83	$-\frac{7}{4}, -\frac{5}{4}, -\frac{1}{2}, -\frac{3}{4}$	1980, 1998	yes
5	7	59, 91, 96, 98, 99, 157, 158	$2, 4, 1, -4, -1, 1, -1$	1036	yes
7	7	112–116, 145, 147	$8, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}$	1157	yes
5	15	0, 2–9, 11, 14, 16–18, 64	whole, up to 23	17	14 of 15

Table 1: Irreducible infeasible sets of held turns on the male body, bounds of separation kept (measured). Turn numbers are those of the first way listed; "bounded arcs": arcs held at length at least one in the support of the non-negative form. Rows without an exact check come from the floating-point linear program; their turns are each closable alone, so a pair is irreducible. In the last row, the whole lengths closing the set without turn 3 were not found within the time limit.

5 Conflicts between turns

The held turns of the body close together if lengths are only asked to be non-negative, not within P_{eng}. Table 1 lists *irreducible* infeasible sets, which cannot be closed within the bounds while every proper subset can; the Farkas multipliers were checked in exact arithmetic, and for each turn left out, whole lengths closing the others were found by HiGHS and checked in exact integer arithmetic (measured).

In four ways the turn round the left wrist cannot even be closed alone: its shift is $-(\ell_{1921} + \ell_{1937})$ modulo the balances, and arc 1937, 0.08 edges long, is held at length at least one (exact for way 0; whether every valid quantization meets this bound we did not check). One may hope that every other conflict contains two parallel turns whose annulus forbids it. The smallest counterexample has three turns near the right eye, every two of which close together: $s_{36} + s_{37} - s_{39} = \ell_{604} + \ell_{619} + \ell_{620} + \ell_{637}$ modulo the balances, with $\ell_{619} \geq 1$. Every set found is infeasible in real numbers already, so none is an obstruction of parity. The conflicts between two turns lie in clusters of singularities at the eyes, hands and feet, the sum or difference of the shifts being a sum of two to eight arc lengths, one bounded; yet in none of the 13 we examined (the pairs of Table 1 but 49, 71 of way 5) are the turns separated by a 3–5 pair on one trace, as in Proposition 3(ii). What separates them is a staircase.

Proposition 6 (Staircases)**.** *Let γ_1, γ_2 be turns and c, c' singular vertices such that*

$$s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \varepsilon_1 d_1 + \varepsilon_2 d_2$$

for all balanced ℓ, with $\varepsilon_1, \varepsilon_2 \in \{+1, -1\}$ and $d = p_{c'} - p_c = d_1 e_1 + d_2 e_2$ the separation of c' from c developed along a path π from c to c' (Propositions 3(i) and 7 give such identities). Let σ be a path from c to c' along arcs $a_1, \ldots, a_r$, through no other singular vertex, and homotopic to π with its ends fixed in Σ without the other singular vertices. Developed along σ, each arc a_i runs along a direction $v_i \in \{\pm e_1, \pm e_2\}$. If $\varepsilon_1 v_{i,1} + \varepsilon_2 v_{i,2} = 1$ for every i, so that σ is a staircase*, using two directions a quarter-turn apart and climbing across the diagonal on which the two turns could close, then*

$$s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \sum_{i=1}^{r} \ell_{a_i}$$

	conflict	between the turns	staircase	mechanism
A	$s_{37} - s_{26}$, ways 1, 3, 7; right eye; disjoint, simple	annulus: 78 (val. 3), 76 (val. 5)	2 arcs, 0.990; held arc 619 by the balances	dislocation
B	$s_{45} - s_{49}$, ways 1, 3; right hand; simple, crossing	lenses: 102, 103 (val. 3)	4 arcs, 1.731; held arc 1069 by the balances	exchange
C	$s_{58} - s_{94}$, ways 5, 7; right hand	109, 118 (val. 3)	3 arcs, 2.203; held arc 858 by the balances	exchange
D	$s_{57} + s_{58}$, way 6; left foot; figure-eights	155, 156 (val. 3)	2 arcs, 1.452, one of them held	exchange
E	$s_{54} - s_{61}$, ways 1, 3, 5, 7; left eye	11 vertices of index 3, and 207 (val. 3)	none; a closed chain of 6 combs	combs
F	$s_8 + s_{15}$, way 7; right foot	14, 23, 24, and 15 (val. 3)	none; a closed chain of 4 combs round a cap	combs

Table 2: The conflicts between two turns on the male body (measured). Turn numbers are those of the first way listed, signs those of the exported shifts; "between the turns": the singular vertices, numbered as nodes, round which the combination of the two curves winds; "staircase": one between them whose length equals the combination modulo the balances, and where the held arc of the certificate lies. A: Propositions 3(i) and 6; B–D: Propositions 7 and 6; E, F: Section 6.1.

for all balanced ℓ. Both turns are rings only if every arc of σ has length zero. If this sum equals, modulo the balances, a form with non-negative coefficients containing an arc held at length at least one, one of the two turns is a helix under every quantization meeting the bounds.

Proof. Developing along σ adds $\ell_{a_i} v_i$ arc after arc, the nodes passed being T-junctions. As σ and π are homotopic with fixed ends away from the other cone points, $d = \sum_i \ell_{a_i} v_i$, so $\varepsilon_1 d_1 + \varepsilon_2 d_2 = \sum_i \ell_{a_i}$, first for positive balanced lengths, then for all by linearity. □

The proof applies to any path on whose steps the form of the shift difference is one; Proposition 3(ii) is the straight case. Exchanges give the other identities.

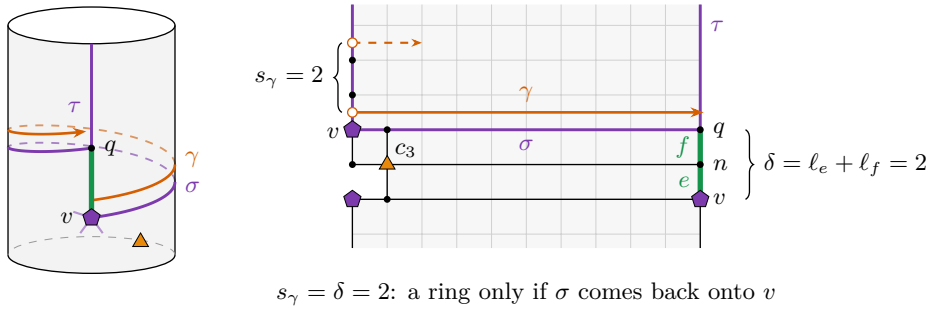
Proposition 7 (Exchanging two vertices)**.** *Let γ_1, γ_2 be turns, and κ a path from the start of $\bar{\gamma}_1$ to that of $\bar{\gamma}_2$ that carries the walking direction of γ_2 onto that of γ_1, as in Proposition 2(b); let $\lambda = \bar{\gamma}_1 \, \kappa \, \bar{\gamma}_2^{-1} \kappa^{-1}$. Suppose that λ is homotopic in Σ° to $\lambda_c \lambda_{c'}^{-1}$, where λ_c and $\lambda_{c'}$ are small loops run with c, c' on their right round two singular vertices of the same valence k, reached along paths that develop them at p_c and $p_{c'}$. Then*

$$s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \langle e_2, \ (I - R^{4-k})(p_c - p_{c'}) \rangle.$$

For valence three or five this is $\pm d_1 \pm d_2$ with $d = p_c - p_{c'}$, as for a dislocation, and Proposition 6 applies.

Proof. As in Proposition 2(b), $\hat{h}(\lambda)$ is the translation by $\hat{h}(\bar{\gamma}_1) - \hat{h}(\bar{\gamma}_2)$. With $j = 4 - k$, $\hat{h}(\lambda_c) \circ \hat{h}(\lambda_{c'})^{-1}(x) = R^j(R^{-j}(x - p_{c'}) + p_{c'} - p_c) + p_c = x + (I - R^j)(p_c - p_{c'})$; for $j = \pm 1$ both components are $\pm d_1 \pm d_2$. □

The 13 conflicts fall into six configurations (Table 2, measured). In A the annulus between two simple turns holds a 3–5 pair, and the staircase joining them, 0.990 long, is the difference of the real shifts; at the engine's whole lengths the pair is one edge apart on one grid line, Proposition 3(ii) with $\delta = 1$ (its hypothesis on κ not checked). In B, C and D the turns cross and their combination winds $+1$ and -1 round two vertices of valence three, an exchange in homology (the homotopy of Proposition 7 was not checked, the staircase identity was). In E and F the combination winds once round vertices of total index four, like a turn round a cap, and no staircase, nor sum of two, was found equal to it.



(a) the cylinder (b) cut open along τ

Figure 3: A separatrix that winds (Proposition 8). (a) On a cylinder such as a wrist, the separatrix σ of the vertex v of valence five goes once round and stops on another separatrix τ of v. (b) Cut open along τ: σ, of length 10, stops at q, at $\delta = \ell_e + \ell_f = 2$ from v, and the turn γ beside σ comes back $2 = \delta$ rows higher; it is a ring only if σ comes back onto v. At the left wrist of the male body $\delta = \ell_{1920} + \ell_{1921}$, kept positive by a bound of separation.

A separatrix that winds. The turn that cannot be closed alone is of another kind (Figure 3).

Proposition 8 (A separatrix that winds)**.** *Let σ and τ be separatrices of a singular vertex v such that σ, followed from v, stops on τ at a point q at distance $\delta(\ell)$ from v along τ, a sum of arc lengths. Let λ be the loop that runs along σ from v to q and back along τ to v, pushed off v into the sector it turns through. If λ is freely homotopic in Σ° to the curve $\bar\gamma$ of a turn γ, the walking direction of γ going to that of σ, then $s_\gamma(\ell) = \pm\delta(\ell)$ for all balanced ℓ. The turn is a ring only if $\delta = 0$, that is if σ comes back onto its own vertex; if δ equals, modulo the balances, a non-negative form containing an arc held at length at least one, γ is a helix under every quantization meeting the bounds.*

Proof. Develop λ from a point of σ next to v, with v at the origin and σ along e_1. The separatrix stops on τ at a right angle, so the return is a segment of length δ along $\pm e_2$. Homotopic to the curve of a turn, λ has a translation as holonomy, which maps v developed at the start onto v developed at the end, at distance δ from q across σ. A free homotopy conjugates the holonomy by a motion whose rotation carries the walking direction of γ onto that of σ, here the identity. □

At the left wrist the turn (42 of way 0, length 6.99, real shift -0.305) bounds a cap of four vertices of valence three, whose Burgers sum gives its shift as a form without sign. The sign comes from outside: the separatrix 964 of the vertex 246, of valence five, runs 6.994 edges once round the wrist and stops on another of its separatrices at $\delta = \ell_{1920} + \ell_{1921} = 0.305$ from 246; the shift is $-\delta$, and ℓ_{1920} equals the length of the held arc 1937, both modulo the balances: whence the certificate and $s \le -1$ at every admissible whole length. We checked the identity, not the homotopy (measured).

Larger conflicts. The three turns at the right eye wind round one 3–5 pair, and their combination is the length of a staircase of three arcs through the held arc 619; the four at the eyes wind round a 3–5 pair and two vertices of valence three, a dislocation and an exchange in homology. The fractional multipliers of Table 1 come from asking the bounded arcs to weigh one in total; at the left hand, where the turns cross each other 10 to 46 times, the primitive ones give

$$-(7s_{66}+5s_{71}+2s_{75}+3s_{76}) = 2F = 10\,\ell_{1805}+2\ell_{1817}+4\ell_{1853}+2(\ell_{1870}+\ell_{1879}+\ell_{1969}+\ell_{1998}+\ell_{2029})$$

modulo the balances. The coefficient 5 is a dislocation: a separatrix of the vertex 231 (valence three) stops on one of 228 (valence five), and arc 1805, 0.063 edges long, joins that junction

to 228, so ℓ_{1805} is the offset of the pair across the line of the field that nearly joins them, Proposition 3(ii) with $\delta = \ell_{1805}$ (hypothesis on κ not checked); their twisted windings (Section 6) contribute exactly $10\,\ell_{1805}$ (measured).

Local dislocations. The curves of an irreducible set need not bound one planar region round a cluster of singularities: configurations B to F and the sets of four and seven turns do not. What the conflicts share is local.

Question 1 (Are conflicts sums of local dislocations?)**.** *On a closed surface of genus zero, let H be an irreducible set of turns that cannot be closed together within the bounds of separation. Does some combination $\sum_{\gamma\in H} u_\gamma s_\gamma$ with whole multipliers that certifies the conflict (Remark 2) equal, modulo the balances, a sum with non-negative whole weights of*

(i) lengths of staircases (Proposition 6) between a vertex of valence three and one of valence five, or between two vertices of equal valence exchanged as in Proposition 7, and

(ii) shifts of caps: sign-definite forms carried by curves that wind once round singular vertices of total index four, as the wrist turn and its separatrix (Proposition 8),

one term at least containing an arc held open?

It holds with one term, up to the unchecked homotopies, for the wrist, configurations A to D and the right eye; Section 6.1 settles it on the body: no as stated, yes and empty when read loosely.

6 Any combination of turns

Winding numbers are the homology class of a weighted cycle $\sum_\gamma u_\gamma \bar\gamma$, which is commutative, while the holonomy is not; the free differential calculus of Fox [13] keeps the order. Let Σ be closed of genus zero, with singular vertices $c_1, \dots, c_n$ (Proposition 14 adds borders and handles). Fix x_0 inside a patch, with its frame, and disjoint paths π_i from x_0 to c_i; let λ_i follow π_i, go round c_i with c_i on its right, and come back. The fundamental group of Σ° is generated by the λ_i with a single relation, their product in the order in which the paths leave x_0. Let p_i be the developed position of c_i at the end of π_i and $R_i = R^{4-k(c_i)}$, so that $\hat h(\lambda_i) = \rho_i$, $\rho_i(x) = R_i(x - p_i) + p_i$. Write $\hat h(\gamma)(x) = \rho(\gamma)x + \tau(\gamma)$; then $\tau(\alpha\beta) = \tau(\alpha) + \rho(\alpha)\,\tau(\beta)$. Let F be the free group on $x_1, \dots, x_n$, mapped onto the fundamental group by $x_i \mapsto \lambda_i$, with the Fox derivatives $\partial x_j/\partial x_i = \delta_{ij}$, $\partial(uv)/\partial x_i = \partial u/\partial x_i + u\,\partial v/\partial x_i$. Extended linearly to the group ring, ρ takes the values $a_0 I + a_1 R + a_2 R^2 + a_3 R^3$, and the augmentation ϵ maps such an element to $a_0 + a_1 + a_2 + a_3$. The first identity below is the defining property of Fox derivatives for a crossed homomorphism [13], and the p_i are, up to a common motion, twisted-cohomology coordinates of the flat metric [33, 34]; what is new is the reading on combinations of turns.

Proposition 9 (Fox–Burgers formula)**.** *For every word w of F,*

$$\tau(w) = \sum_{i=1}^{n} \rho\Big(\frac{\partial w}{\partial x_i}\Big)(I - R_i)\,p_i.$$

Let $\gamma_1, \dots, \gamma_m$ be turns, κ_r a path from x_0 to the start of $\bar\gamma_r$, f_r the direction e_2 of γ_r carried back to x_0 along κ_r (one of $\pm e_1$, $\pm e_2$), and w_r a word for $\kappa_r \bar\gamma_r \kappa_r^{-1}$. For multipliers u_r define the twisted winding *of the weighted cycle round c_i,*

$$W_i = \sum_{r=1}^{m} u_r\, \rho\Big(\frac{\partial w_r}{\partial x_i}\Big)^{\top} f_r \in \mathbb{Z}^2 \ \textit{(for whole } u_r\textit{)}.$$

Then, for all balanced lengths,

$$\sum_{r=1}^{m} u_r\, s_{\gamma_r}(\ell) = \sum_{i=1}^{n} \langle W_i,\ (I - R_i)\, p_i \rangle, \qquad \sum_{i=1}^{n} (I - R_i)^\top W_i = 0,$$

so the sum does not change when all the p_i are moved by one translation: it is a combination of separations $p_i - p_j$. The ordinary winding number of $\sum_r u_r \bar{\gamma}_r$ round c_i is, up to a constant common to all i, $w_i = \sum_r u_r\, \epsilon(\partial w_r / \partial x_i)$. The words w_r are defined up to the relation, and the W_i accordingly: inserting a conjugate $g \varrho g^{-1}$ of the relation ϱ in w_r, after a prefix a of w_r, adds $u_r \rho(ag\, \partial \varrho / \partial x_i)^\top f_r$ to every W_i (with a empty, $u_r \rho(g\, \partial \varrho / \partial x_i)^\top f_r$), and leaves the sum unchanged.

Proof. Both $w \mapsto \tau(w)$ and the right-hand side satisfy $\phi(uv) = \phi(u) + \rho(u)\phi(v)$, and they agree on the letters and the empty word, so they agree everywhere by induction on the length (for x_i^{-1}, use $0 = \phi(x_i x_i^{-1})$). For a turn, $\hat{h}(\kappa_r \bar{\gamma}_r \kappa_r^{-1})$ is the translation by $M_r T_r$, with T_r the translation of $\bar{\gamma}_r$ in its own frame and M_r the rotation of $\hat{h}(\kappa_r)$; so $s_{\gamma_r} = \langle M_r e_2, M_r T_r \rangle = \langle f_r, \tau(w_r) \rangle$, and $\langle f, \rho(D) b \rangle = \langle \rho(D)^\top f, b \rangle$ gives the formula. The fundamental formula $\sum_i (\partial w / \partial x_i)(x_i - 1) = w - 1$ maps under ρ to $\sum_i \rho(\partial w_r / \partial x_i)(R_i - I) = \rho(w_r) - I = 0$, the rotation of a turn being trivial; transposed and summed with the $u_r f_r$, it is the second identity. Finally $\epsilon(\partial w / \partial x_i)$ is the exponent sum of x_i in w, the homology class of w on the basis of the λ_i, defined up to the relation, in which each λ_i appears once. □

For a simple turn $\rho(\partial w / \partial x_i) = Q_i$ and the formula is Proposition 2(a). In general a cycle may go round c_i several times after different rotations; modulo 2 these disappear.

Corollary 2 (Checkerboard rule for any combination)**.** *On a closed surface of genus zero, let ℓ be whole and balanced and the multipliers u_r whole. Colour each singular vertex c_i by the parity χ_i of $p_{i,1} + p_{i,2}$; it does not depend on the paths π_i, and for positive lengths it is the colour of c_i in the bipartition of the quad mesh. Then*

$$\sum_r u_r\, s_{\gamma_r}(\ell) \equiv \sum_r u_r\, \alpha_{\gamma_r}(\ell) \equiv \sum_{i:\ k(c_i)\ odd} w_i\, \chi_i \pmod 2,$$

with w_i the winding numbers of $\sum_r u_r \bar{\gamma}_r$, whatever the common constant and whatever the orientation of each $\bar{\gamma}_r$. In particular, a combination whose cycle winds an even number of times round every vertex of odd valence takes even values at every whole length.

Proof. Since $(R-I)^2 = -2R$, $R^k(I - R^{\pm 1}) \equiv I - R^{\pm 1}$ modulo 2, so $\rho(D)(I - R_i) \equiv \epsilon(D)(I - R_i)$ for $k(c_i)$ odd, and $\equiv 0$ for $k(c_i)$ even, where $I - R_i \in \{0, 2I\}$; and $\langle f, (I - R^{\pm 1})p \rangle \equiv p_1 + p_2$ for $f \in \{\pm e_1, \pm e_2\}$. This gives the sum of Proposition 9 modulo 2, and the direction e_1 of each turn that of $\sum_r u_r \alpha_{\gamma_r}$. Rotations about whole points and quarter-turns keep $p_1 + p_2$ mod 2, so χ_i does not depend on the paths. A whole translation of the origin flips all colours and adds $\sum_{k(c_i)\ \mathrm{odd}} w_i$, even because the rotation of the cycle is trivial ($\sum_i w_i(4 - k(c_i)) \equiv 0 \pmod 4$); adding q to all w_i adds $q \sum_{k(c_i)\ \mathrm{odd}} \chi_i$, the value modulo 2 of the relation, a loop with zero translation, hence even. Reversing a turn changes signs only. For positive lengths the developed parity colours the quad graph properly, as in Proposition 5. □

The corollary contains Propositions 4, 3(iii) and 5. As for crystal defects [26] and in [19, §7], winding numbers do not determine the holonomy:

Proposition 10 (Winding numbers do not suffice)**.** *Let c_a, c_b be singular vertices of odd valence developed at distinct points $p_a \neq p_b$. The loop $\lambda_a \lambda_b \lambda_a^{-1} \lambda_b^{-1}$ winds zero times round every singular vertex, and its holonomy is the translation by $(I - R_a)(I - R_b)(p_a - p_b)$, of length $2|p_a - p_b| \neq 0$. No function of the winding numbers gives the holonomy of the loops with trivial rotation.*

Proof. For $w = x_a x_b x_a^{-1} x_b^{-1}$, $\rho(\partial w / \partial x_a) = \rho(1 - x_a x_b x_a^{-1}) = I - R_b$ and $\rho(\partial w / \partial x_b) = \rho(x_a - w) = R_a - I$, the rotations commuting; Proposition 9 gives the translation, and for odd valences $I - R_a$, $I - R_b$ are $\sqrt{2}$ times rotations. The exponent sums of w are zero. □

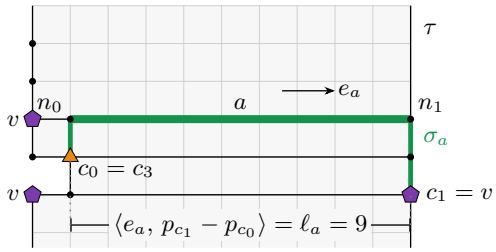


Figure 4: The comb σ_a of an arc a (Lemma 3), $\ell_a = 9$: from the foot c_0 of one end of a, of valence three, up its trace, along a, and down the other perpendicular trace to the foot c_1, of valence five. Every step but a is orthogonal to a, so $\langle e_a, p_{c_1} - p_{c_0} \rangle = 9 = \ell_a$. This T-mesh is that of Figures 2(b) and 3(b): the staircase, the winding separatrix and the comb are three readings of the same shift 2.

On the male body. Cut into one chart along a spanning tree of the nodes, the T-mesh gives every turn a word, and Proposition 9 its translation, for all 1,299 turns. Corollary 2 holds on all of them and on the 14 sets of Table 1 with whole multipliers; at the left hand the cycle winds an even number of times round every vertex of odd valence, which is why $-2F$ is even. Four turns satisfy $w_{18} = w_{22} + w_{101} + w_{65}$ round every singular vertex while their translations do not, in any quarter-turn frames: a "winding Burgers formula" with numbers for weights is false (measured).

6.1 Combs

The twisted windings depend on the chart; combs do not. In the T-meshes of openclare, traced from the separatrices of the singular vertices only, every trace of a closed surface starts at a singular vertex (feature lines or seeded traces would break this; we assume it). For an arc a, along e_a from the node n_0 to the node n_1, the *foot* of n_j is the node itself if it is singular, and otherwise the singular vertex where the trace through n_j perpendicular to a starts. We assume this vertex unique: a single trace perpendicular to a meets n_j (not two, from opposite sides), and it is neither a separatrix connection nor a loop; otherwise a foot must be chosen, and different choices give different dislocations. The *comb* σ_a of a runs from the foot c_0 of n_0 along its trace to n_0, along a to n_1, and along the other trace to the foot c_1 of n_1, through T-junctions only (Figure 4). The *graph of feet* has the singular vertices of odd valence as vertices and the arcs whose two feet are distinct vertices of odd valence as edges.

Lemma 3 (Combs)**.** *Let p_{c_0}, p_{c_1} be the feet of an arc a developed along its comb. Then, for all balanced lengths,*

$$\langle e_a,\ p_{c_1} - p_{c_0} \rangle = \ell_a.$$

If $c_0 \neq c_1$ have odd valence, $2\ell_a$ is, up to sign, the form of the twisted class $\lambda_{c_0}\lambda_{c_1}^{\pm 1} \otimes v$ of the loops round the two feet reached along the comb, for a whole v with $(I - R^{\pm 1})^\top v = 2e_a$: the pair is a dislocation if the valences differ (Proposition 3) and an exchange if they are equal (Proposition 7). If $c_0 = c_1$, ℓ_a is the shift of a winding separatrix (Proposition 8).

Proof. Every step of the comb but a runs along a trace perpendicular to a, through no cone point, and adds a vector orthogonal to e_a; the step along a adds $\ell_a e_a$. The loop $\lambda_{c_0}\lambda_{c_1}^{\pm 1}$, with the sign that makes its rotation trivial, has the translation $\pm(I - R^{\pm 1})(p_{c_0} - p_{c_1})$ as in Propositions 3 and 7; pairing it with v gives $\pm 2\ell_a$, and v is whole because $I - R^{\pm 1}$ maps $\mathbb{Z}^2$ onto the vectors whose coordinates have equal parity, among them $2e_a$. If $c_0 = c_1$, the second trace is a separatrix of the same vertex stopping on the first at the distance ℓ_a. □

A *closed walk of combs* is a sequence of arcs $a_1, \ldots, a_k$ whose feet are c_{i-1} and c_i, of odd valence, with $c_k = c_0$; its *reading* is $\sum_i \ell_{a_i}$. A walk of length one is a winding separatrix, one of length two between valences three and five the staircase of Proposition 6, and one round four vertices of valence three a cap.

Proposition 11 (Certificates are sums of combs)**.** *Let $c = \sum_a c_a \ell_a$, $c_a \in \mathbb{Z}_{\geq 0}$, be a certificate (Remark 2) whose arcs have feet of odd valence and whose odd part, the arcs with c_a odd, is a cycle modulo 2 of the graph of feet. Then c is a sum with whole non-negative weights of comb readings $2\ell_a$, each the form of a dislocation or an exchange (Lemma 3), and of readings of closed walks of combs. With real weights, every non-negative form supported on arcs whose two feet are of odd valence, distinct or equal, is a non-negative combination of comb readings $2\ell_a$ and of winding separatrices ℓ_a, without condition of parity.*

Proof. The odd part splits into closed walks along its edges (Veblen), and what remains is $\sum_a \lfloor c_a/2 \rfloor \, 2\ell_a$. With real weights, $c = \sum_a \frac{c_a}{2} \, (2\ell_a)$ on the arcs with two distinct feet, and $c_a \ell_a$ on those whose feet are one vertex. □

Remark 4 (Closed walks as twisted cycles)**.** *We expect the reading of a closed walk of combs, $\sum_i \langle e_{a_i} - e_{a_{i+1}}, \; p_{c_i} \rangle$ by a summation by parts, to be the Burgers form of a twisted cycle near its combs; but the two combs through c_i develop it in charts that differ by a rotation, and a proof must develop the whole walk in one chart. We use closed walks only as forms.*

Proposition 12 (The Burgers pairing)**.** *Let the surface be closed of genus zero, with no singular vertex of valence divisible by four, and let every trace of the T-mesh start at a singular vertex. Then the map that sends a class of the twisted homology $H_1(\Sigma^\circ; \mathbb{R}^2_\rho)$ to its form $\sum_i \langle W_i, (I - R_i) p_i \rangle$ on the balanced lengths is onto the dual of the balanced lengths. Its kernel contains the class across each separatrix connection. If some singular vertices have valence divisible by four, no form depends on their developed positions, and the map need not be onto.*

Proof. If every such form vanishes at balanced ℓ, every loop of Σ° with trivial rotation has a holonomy without translation (Proposition 9). The commutator of λ_a and λ_b has trivial rotation and translation $(I - R_a)(I - R_b)(p_a - p_b)$ (Proposition 10); no R_i is the identity, so $I - R_i$ is invertible ($\sqrt{2}$ times a rotation, or $2I$), and every singular vertex develops at one point q. As the ρ_i fix q and generate the holonomy on a closed sphere, every singular vertex develops at q along any path; every arc has feet, so every comb offset of Lemma 3 is zero, and $\ell = 0$. The image has no common zero but 0. Across a connection the separation of its two vertices is zero across the trace, and the twisted class pairing the across direction with it has the zero form. A vertex of valence divisible by four has $(I - R_i) p_i = 0$: its position is a relative period, which the lengths can move without changing any holonomy. □

On the male body. Lemma 3 holds on all 2,069 arcs, 2,046 of which have two distinct feet of odd valence: nearly every arc is a local dislocation or exchange. The twisted homology has dimension $518 = 2(n - 2)$ and the pairing rank 517, its kernel the class across the one separatrix connection; we conjecture that without vertices of valence divisible by four the kernel is spanned by the classes across the connections. All 25 certificates of the conflicts examined have a representative whose odd part is a cycle of the graph of feet (exact). The wrist is a comb with both feet at 246, the winding separatrix; A, C and D are closed walks of two combs (A the staircase $\ell_{611} + \ell_{624}$), B of four, the right eye of three, E of six, of total index -2, and F of four round a cap; the left hand is five times the dislocation ℓ_{1805} plus eight comb readings on a tree of seven arcs, and the set of seven of way 7 a tree of seven combs (measured).

Question 1, read loosely, is true and empty. Allowing every comb reading and every closed walk of combs, Proposition 11 answers it for every certificate whose odd part is a cycle of the graph of feet, all those of the body; but this follows from Farkas' lemma and a parity, and says nothing about which configurations conflict. *Read strictly, it is false*: restricted to staircases of weight one, monotone walks round a cap and winding separatrices, exact rational balanced lengths make every such form non-negative and the certificates of E and F negative, -2.9967 and -1.9973 (checked in integer arithmetic). These forms are infinitely many, but each

is the length of a walk on a graph whose steps carry the arc lengths, so a shortest-path search finds a negative one if any. In floating point the same test also separates the four turns at the eyes and puts the left hand inside the cone (measured). Whether every certificate splits into combs on disjoint sets of vertices, each of its sign, is open: the trees of the left hand and of way 7 share vertices.

7 Borders and handles

Models with borders (the eye sockets of Suzanne, the mouth of Jerry) and handles (Bob, a torus; part and cage) need the forms below, measured on the T-meshes of these bench models with an engine version not recorded. A border node b is a corner of angle $a(b)\pi/2$, of *index* $2-a(b)$; for a border component β_j, $L_j(\ell)$ is the sum of the lengths of its arcs.

Lemma 4 (Gauss–Bonnet with borders)**.** *Let D be a region bounded by curves of turns and by border components. Then*

$$\sum_{c\in D}(4-k(c))+\sum_{b\in D\cap\partial\Sigma}(2-a(b))=4\chi(D)-\Theta,$$

where Θ is the turning of the curves of the turns at their corners, in quarter-turns: zero for the side of a simple curve.

Proof. Gauss–Bonnet for a flat surface with cone points and a piecewise straight boundary, border nodes being corners of turning $(2-a)\pi/2$. □

A side of a simple turn holding b border components without corner and h handles has index $4(1-b-2h)$, and may hold no singular vertex at all; the lemma holds on the 1,076 sides of turns of Suzanne and Jerry (measured). For positive lengths the homomorphism of the next theorem is the cycle parity of the quad graph, a $\mathbb{Z}/2$-homology invariant of any quadrangulation [21, Lemma 3.1] extended here to turns at all balanced lengths.

Theorem 1 (Parity of a turn on any surface)**.** *Let ℓ be whole and balanced. There is a homomorphism $\Omega_\ell\colon H_1(\Sigma;\mathbb{Z}/2)\to\mathbb{Z}/2$ such that for every turn*

$$\alpha_\gamma(\ell)+s_\gamma(\ell)\equiv\Omega_\ell([\bar\gamma])\pmod 2.$$

Its value on a border component is the parity of $L_j(\ell)$, the sum of all $L_j(\ell)$ is even, and for positive lengths $\Omega_\ell(z)$ is the parity of the length of any cycle of the quad graph in the class z; $\Omega_\ell=0$ exactly when the quad graph is bipartite. On a surface of genus zero with borders, $\alpha_\gamma+s_\gamma$ has the parity of the total length of the border components on either side of $\bar\gamma$, or of the number of times $\bar\gamma$ crosses any system of disjoint arcs joining the border components of odd length in pairs.

Proof. As in Proposition 5, $\gamma\mapsto\pi(\hat h(\gamma))$ is a homomorphism, zero on the loops round the singular vertices (quarter-turns about whole points); with values of exponent two it factors through $H_1(\Sigma;\mathbb{Z}/2)$, those loops spanning the kernel of $H_1(\Sigma^\circ)\to H_1(\Sigma)$. For a turn $\hat h(\bar\gamma)$ is $(\alpha_\gamma,\pm s_\gamma)$ up to a quarter-turn; a border component comes back displaced by $\sum_a\pm\ell_a v_a$, of the parity of $L_j(\ell)$, and the components add up to the boundary, zero in homology. Along a cycle of the quad graph every edge changes the colour. In genus zero the border components span $H_1(\Sigma;\mathbb{Z}/2)$ and $\bar\gamma$ is homologous to those on one side; the last form is Lefschetz duality. □

On Suzanne, whose eye sockets are borders of 23 and 25 edges, the turn round an eye socket is a ring of 23 quads at the lengths of one way and of 24 at another, against Proposition 5; its shift is zero at every balanced length:

Proposition 13 (Collars)**.** *Let the curve of a turn γ and a border component β_j without corner bound an annulus holding no singular vertex. Then $s_\gamma(\ell) = 0$ and $\alpha_\gamma(\ell) = L_j(\ell)$ for all balanced lengths.*

Proof. The holonomy of β_j is the translation by L_j along an axis of the field. Cut along a path from $\bar\gamma$ to β_j, the annulus is a disc without cone point, so the translation of $\bar\gamma$ is $L_j u$ with $u \in \{\pm e_1, \pm e_2\}$, as in Proposition 2(b). For positive lengths $\alpha_\gamma > 0$, so $u = \pm e_1$; the identities are linear and extend to all balanced lengths. □

The theorem predicts the parity of all 660 turns of Suzanne, Jerry and a dome, 61 of them odd, and of every closed loop of the output meshes (294 on the male body); the proposition holds on all 40 collars; on Bob, Ω_ℓ depends on the lengths, the three ways giving it $(0,1)$, $(0,0)$ and $(1,0)$ on two handle loops (measured). For combinations, $\sum_r u_r(\alpha_{\gamma_r} + s_{\gamma_r}) \equiv \sum_r u_r\, \Omega_\ell([\bar\gamma_r])$; the colours of Corollary 2 may depend on the paths, but the count on white points still gives the parity when every curve lies in a planar region without border.

Proposition 14 (Burgers sum with borders and handles)**.** *Let the loops λ_i round the singular vertices, β_j round the border components and a_k, b_k round the handles generate the fundamental group of Σ°. For every word w in them,*

$$\tau(w) = \sum_i \rho\Big(\frac{\partial w}{\partial x_i}\Big)(I - R_i)\, p_i + \sum_{g \in \{\beta_j, a_k, b_k\}} \rho\Big(\frac{\partial w}{\partial g}\Big)\tau(g).$$

For a turn, the shift is the Burgers sum of the singular vertices plus a period term*, linear in the lengths and not determined by the developed positions of the singular vertices. A border component without corner has holonomy the translation by L_j along its developed direction; a handle inside the region between two turns contributes the commutator $(I - \rho(b))\tau(a) - (I - \rho(a))\tau(b)$, zero when both its loops have trivial rotation.*

Proof. Both sides satisfy $\phi(uv) = \phi(u) + \rho(u)\phi(v)$ and agree on the generators. The commutator of $x \mapsto Ax + a$ and $x \mapsto Bx + b$ with commuting rotations is the translation by $(I - B)a - (I - A)b$. □

With its border term the sum reproduces all 314 sides holding a border on which it was tested, and without it none of the 586 such sides of Suzanne and Jerry (measured).

Proposition 15 (Two crossing turns on a torus)**.** *On a closed surface of genus one, let the curves of two turns be simple and cross once. Cut along both, the surface is a disc D holding every singular vertex, and its Burgers sum vanishes: $\sum_i Q_i(I - R_i)p_i = 0$, in the order of first visit, for all balanced lengths. More generally, if κ is a simple loop crossing the curve of a turn once, with rotation R^j, then $(I - R^j)\, T_\gamma = \pm B_D$: for $j \neq 0$ the translation of the turn is set by the singular vertices, for $j = 0$ it is a free period and the singular vertices satisfy $B_D = 0$.*

Proof. The border of D is the commutator of the two curves, whose holonomies are translations and commute; Proposition 2(a) applies to D. With κ of rotation R^j, the commutator is the translation by $(I - R^j)T_\gamma$. □

On Bob, of genus one with ten vertices of valence three and ten of valence five, the Burgers sum vanishes for all 304 pairs of simple turns crossing once, and for none of 90 shuffled orders (measured). With a single 3–5 pair the two cone points would develop onto one point; that no such torus exists is the theorem of Barnette, Jucovič and Trenkler [3], also a consequence of Abel's theorem ($c_5 - c_3$ is principal on an elliptic curve only if $c_5 = c_3$), an obstruction to existence, not to closing. With $j = 0$ the proposition is the commutator argument of [19, §7]. The conflict met on Bob is of another kind: $3s_{33} - 2s_{31}$, for two turns winding three and two

times round the same handle, cancels the period and equals two arc lengths at a vertex of valence three, one held open (measured).

Proposition 1, Lemma 1 for discs and annuli, Propositions 2 to 4 (for annuli), 6, 7, 8 and 10, Remarks 2 and 3, Theorem 2 and Corollary 3 need no closed sphere; Lemma 2, Proposition 5, Corollary 1, the count of independent rings, the setting of Proposition 9, Corollary 2 and Question 1 do; Appendix A treats triangle meshes.

8 Turns every quantization within bounds leaves shifted

Turned around, the obstructions bound from below the number of turns any quantization in a set P of lengths leaves shifted.

Theorem 2 (Forced shifts)**.** *Let H be a finite set of turns of a T-mesh and P a non-empty set of lengths containing every quantization considered: for instance the balanced lengths meeting given bounds of separation, $\ell \geq b$, possibly with a cap $\frac{1}{n}\sum_a |\ell_a - t_a| \leq D$ on the mean deviation from the targets t, or with a given number of quads. Let*

$$\tau_P(H) = \min\{\, |X| : X \subseteq H,\ \textit{the turns of } H \setminus X \textit{ can be closed together by lengths in } P \,\}.$$

Then for every $\ell \in P$, whole or not, at least $\tau_P(H)$ turns of H have $s_\gamma(\ell) \neq 0$. If H contains k pairwise disjoint subsets none of which can be closed within P, then $\tau_P(H) \geq k$.

Proof. At $\ell \in P$ the turns with zero shift are closed by ℓ itself, so the shifted ones form a set X as in the definition. A set whose removal leaves a closable set meets each of the k disjoint subsets. □

The theorem is elementary; its content is in what makes subsets infeasible. It uses neither convexity, nor linearity, nor whole lengths. The minimal non-closable sets are irreducible infeasible subsystems (IIS), $\tau_P(H)$ is a minimum IIS cover [2, 30], and the theorem is the packing bound of that covering problem.

Corollary 3 (Chains of dislocations)**.** *Let $\gamma_0, \gamma_1, \ldots, \gamma_m$ be turns and suppose that for $i = 1, \ldots, m$*

$$s_{\gamma_{i-1}}(\ell) - s_{\gamma_i}(\ell) = \varepsilon_i\, F_i(\ell) \quad \textit{for all balanced } \ell,$$

with $\varepsilon_i \in \{+1, -1\}$ and forms F_i that are positive on the whole points of P; for instance F_i the length of a staircase (Proposition 6) between a vertex of valence three and one of valence five lying between γ_{i-1} and γ_i (Propositions 3 and 2(b)), the staircase containing an arc held at length at least one. Let q be the number of indices $i < m$ with $\varepsilon_{i+1} \neq \varepsilon_i$. Then for every whole $\ell \in P$ at most $q + 1$ of the turns γ_i are rings: at least $m - q$ are helices. In particular $m + 1$ nested parallel turns separated by m dislocations whose staircases all climb the same way leave at least m helices.

Proof. If γ_i and γ_j, $i < j$, are rings, $\sum_{k=i+1}^{j} \varepsilon_k F_k(\ell) = 0$ with $F_k(\ell) > 0$, so a change of sign lies between i and j; between two consecutive rings lies one of its own. □

Remark 5 (What the bound counts)**.** *The theorem counts turns with a non-zero shift; at lengths where a turn is not a strip of the mesh [14, Prop. 2], or where one spiral strip carries several turns of H, it is not a count of helices.*

The bound is attained, by the standard correspondence between minimum IIS covers and maximum feasible subsystems [2, 30]:

Theorem 3 (Forced shifts as a transversal number; standard)**.** *Call $Y \subseteq H$* closable *in P if some $\ell \in P$ has $s_\gamma(\ell) = 0$ for all $\gamma \in Y$, and a* core *a set that is not closable while all its proper subsets are. For any P: (i) $H \setminus X$ is closable if and only if X meets every core; hence $\tau_P(H)$ is the least size of a set meeting every core; (ii) for every $\ell \in P$ the turns ℓ leaves shifted meet every core; (iii) every least such set is exactly the set of turns left shifted by some $\ell \in P$; (iv) $P \subseteq P'$ implies $\tau_P(H) \geq \tau_{P'}(H)$.*

Proof. Closability is hereditary: if X misses a core, $H \setminus X$ contains it, and a non-closable $H \setminus X$ contains a core. The zero-shift set of ℓ is closed by ℓ. For a least X, a length closing $H \setminus X$ shifts a subset of X meeting every core, hence all of X. □

So $\tau_P(H)$ lies between the largest number of disjoint cores and the least hitting set of the cores found so far, exact once its complement is closable; the *implicit hitting-set* loop of Parker and Ryan [30] alternates the two, with cores from the deletion filter of [8].

8.1 Whole lengths and size

Proposition 16 (Whole lengths and size without a cap)**.** *If P is given by the balances and bounds $\ell \geq b \geq 0$, a set closable in real lengths is closable in whole lengths, and closable in real lengths at any number of quads greater than $\inf_{P_Y} Q$, and at $\inf_{P_Y} Q$ itself when the infimum is attained, where P_Y is the set of lengths of P closing Y and $Q(\ell)$ the number of quads.*

Proof. Scale a rational point of P_Y by its denominator (Remark 3). For a size q, take $\ell \in P_Y$ with $0 < Q(\ell) \leq q$; $c\ell \in P_Y$ for $c \geq 1$, and $Q(c\ell) = c^2 Q(\ell)$ reaches q. □

With a cap on the deviation, P_Y is a polytope and $Q(P_Y)$ an interval: two exact points on either side of a window of sizes certify that Y closes within it, while the converse needs the extremes of Q over a polytope. The cap is a budget shared by all turns: on the body the lengths may move by 980.7 edges in all, while closing the 109 strips of way 2 that the bounds do not force costs 1,680.2, and the cores become covers of a knapsack (measured).

On a closed surface where no patch borders itself, an arc lies on a side of each of the two patches it borders, so its column of B has two entries ± 1: B is the incidence matrix of a *bidirected graph* G whose nodes are the balances and whose edges are the arcs, an arc on a border giving a half-edge, the setting of [17], with the rank of Zaslavsky [36]. A cycle of G is *balanced* if values ± 1 on its edges balance each of its nodes, and *unbalanced* otherwise.

Lemma 5 (Handshake)**.** *Let $x \in \mathbb{Z}^n$ and C be a connected component of G. Then $\sum_{r \in C} (Bx)_r \equiv \sum_a x_a \pmod 2$, the second sum running over the border arcs of C. On a closed surface, $\sum_r (Bx)_r$ is even.*

Proof. It is $\sum_a x_a \sum_{r \in C} B_{ra}$, and each column sums to 0 or ± 2, or to ± 1 for a border arc. □

So on a closed surface a system $Bx = d$ with $\sum_r d_r$ odd has real solutions but no whole one; this does not affect the balanced lengths:

Proposition 17 (Free arcs)**.** *Let G be connected and contain an unbalanced cycle. Let T be the edges of a spanning tree of G and one more edge closing an unbalanced cycle, and F the other edges. Then B has full row rank, and every $x_F \in \mathbb{Z}^F$ extends to exactly one $x \in \mathbb{Z}^n$ with $Bx = 0$. The whole balanced lengths thus form a lattice isomorphic to $\mathbb{Z}^F$, of dimension $n - m$ for m balances, with a basis of vectors z_f, $f \in F$, each supported on $T \cup \{f\}$ with $z_f(f) = 1$ and entries in $\{0, \pm 1, \pm 2\}$.*

Proof. T exists: the sign of a cycle is a homomorphism on the cycle space over $\mathbb{Z}/2$, which the fundamental cycles of a spanning tree span, so if they were all balanced every cycle would be. T has m edges. If $B_T y = 0$, a node met by a single edge of T forces that edge to zero; pruning

way	turns	bounds	bounds, cap	cap, size, whole	left shifted by the engine
0	48	1	1	1	13
1	78	5	5	5–6	39
2	52	1	1	1	16
3	86	4	4	5–18	44
4	84	1	1	1–12	49
5	160	6	7–17	7–86	115
6	88	1	1	1–17	44
7	169	9	9–26	9–90	123
2, strips of the mesh	111	2	3–11	3–43	81

Table 3: Male body: turns of each family that every quantization in P leaves shifted (Theorems 2 and 3; measured), with $P = P_{\mathrm{eng}}$, then also the engine's cap (real lengths), then also whole lengths and the engine's window of sizes. A single number is exact; a range is certified, below by the solver's proven bound or disjoint cores, above by a closable subset found within the time limit (120 to 1,800 seconds). Last column: turns of the family shifted at the engine's whole lengths. Turns are counted up to sign.

such edges leaves the unbalanced cycle, on which each value fixes the next up to sign, and going round forces all to zero. So B_T is invertible, B has full row rank, and x_F has exactly one real extension. For $f \in F$, $T \cup \{f\}$ has a one-dimensional space of balanced values. Pruned, it is either three paths between two nodes, whose cycles are the unbalanced cycle of T, one more unbalanced and one balanced (the number of unbalanced cycles of such a graph is even), the balanced one containing f, since the cycle of T does not, and carrying values ± 1; or two cycles joined by a path, possibly of length zero, the cycle through f being balanced and carrying ± 1, or unbalanced, and then ± 1 on both cycles and ± 2 on the path are balanced. Either way there is a whole balanced z_f supported on $T \cup \{f\}$ with $z_f(f) = \pm 1$; with signs fixed, $x = \sum_{f \in F} x_f z_f$ is the whole extension. □

With a spanning forest and, in each component with an unbalanced cycle, an edge closing one or a half-edge, the proposition extends beyond the closed sphere, where full row rank can fail (on part, 145 of 148). On the male body G is connected with an unbalanced cycle and the lattice of balanced lengths has dimension $2{,}069 - 1{,}552 = 517$: balances alone never obstruct whole lengths, and integer elimination ends with a single row of greatest common divisor 2. The shifts of the held turns leave lattices of dimension 363 to 469, and in ways 4, 5 and 6 elimination needs steps of Euclid's algorithm: turn rows can destroy the free-arc structure (measured). The one integer obstruction found on the body is a rounding gap in deviation, not a parity: rings that close in real numbers at a mean deviation of 0.399 need at least 0.536 in whole numbers [14, §7].

8.2 On the male body

In Table 3, H is the held turns of a way, or the 111 strips of the engine's mesh for way 2 leaning less than 0.12, and P is P_{eng}, then also the engine's cap $D = 0.474$ on the mean deviation, then also whole lengths and its window of 10,379 to 10,801 quads. With the bounds alone the implicit hitting-set loop is exact in at most 66 seconds per family; with the cap and the window, bounds come from closable subsets, disjoint cores, solver bounds and whole points checked exactly.

With the bounds alone the forced turns are exactly 1, 5, 1, 4, 1, 6, 1 and 9 (Table 3); in ways 0, 2, 4 and 6 the forced turn is the wrist; in way 7 nine turns are needed although at most eight of its 21 cores are disjoint, so the packing bound is not always attained. With the cap and the window, at whole lengths, the value is still exactly 1 in ways 0 and 2: whole lengths closing every held turn but the wrist exist at 10,439 and 10,436 quads, at mean deviations 0.442 and 0.469. In way 3 integrality costs at least one turn (4 in real numbers, at least 5 in whole

numbers). Of the 16 held turns of way 2 the engine leaves shifted, the bounds force 1, the cap, the size and integrality none: the heuristic accounts for 15; in the other ways it leaves at least 12 to 37 turns more than the best lengths found. The cap is the engine's only approximately (it refused 0.4737), and optima that bring other arcs to zero can break the mesh; keeping open every arc the engine keeps open, the optimum for way 2 is 4 (measured).

9 Choosing the turns

Call Ring-Closing the problem: given the balances B, bounds b, a set H of linear forms (the shifts) and a number k, are there lengths in $P = \{\ell : B\ell = 0,\ \ell \geq b\}$ that close at least k forms of H? The maximum feasible subsystem problem is NP-hard and hard to approximate in general [1], and minimum IIS covers are strongly NP-hard already for network flows [20]; the hardness survives the structure of quantization, a case new as far as we found.

Theorem 4 (Choosing the turns is NP-hard)**.** Ring-Closing *is NP-complete, in real or in whole lengths, already when B is the incidence matrix of a bidirected graph, the bounds are* 0 *or* 1*, and the forms have coefficients* 0 *or* 1*; it is strongly NP-hard, and stays NP-hard with a cap on the deviation, which may be loose. Moreover $\tau_P(H)$ can equal the vertex cover number of any graph, with $|H|$ its number of vertices. So it cannot be approximated within $\sqrt{2} - \varepsilon$ unless $P = NP$ [10, 11, 24, 25], nor within $2 - \varepsilon$ if the unique games conjecture holds [23], and the largest closable subset of H cannot be approximated within $|H|^{1-\varepsilon}$ unless $P = NP$ [16, 37].*

Proof. Membership: closing a given set of k forms is a linear program, and by Proposition 16 real and whole closability agree. Hardness, from Independent Set: for a graph G, take for each edge $e = uv$ four arcs $y_{u,e}$, $y_{v,e}$, d_e, d'_e with the balances

$$y_{u,e} + y_{v,e} - d_e = 0, \qquad y_{u,e} + y_{v,e} - d'_e = 0, \qquad d_e - d'_e = 0,$$

in which every arc appears in exactly two rows with coefficients ± 1, the bounds $d_e, d'_e \geq 1$ and $y \geq 0$, and for each vertex u the form $s_u = \sum_{e \ni u} y_{u,e}$. With $y \geq 0$, $s_u = 0$ forces $y_{u,e} = 0$ at u, so closing two adjacent vertices gives $d_e = 0 < 1$. For an independent set Y, every edge has at most one end in Y: put $y = 0$ on its side and 1 on the other, or, with no end in Y, 1 on either side and 0 on the other; then $d_e = d'_e = 1$, whole lengths in P closing Y. The closable sets are the independent sets of G, and $\tau_P(H) = |V| - \alpha(G)$ is the vertex cover number. The instance is linear in G, with numbers 0 and ± 1; a large cap constrains nothing. □

The reduction was checked by computer on 40 random graphs (measured); its third balance puts every arc in two balances, as in a T-mesh.

Conjecture 1 (Realizable instances)**.** Ring-Closing *stays NP-hard when B, b and H are the balances, the bounds of separation and the turns of a T-mesh of a closed surface of genus zero.*

The evidence is that vertex cover stays NP-hard on planar graphs of small degree [15], and that two turns separated by a dislocation conflict like the two ends of an edge.

Some parts are easy: without turns and window, whole quantization is a polynomial matching problem [12]; with disjoint known cores $\tau_P(H)$ is their number; with shift forms of fixed rank the closable sets are the flats of an arrangement, but on the body the held turns of a way have full rank.

Remark 6 (The window of sizes; a sketch)**.** *The window on the number of quads makes even closing nothing hard in the weak sense. Take a disjoint union of tori, each of two patches whose sides have the targets Nv_i and* 1*, so that $Q = \sum_i 2a_i c_i$, and a cap that lets the short sides c_i grow by T_0 edges in all. For N large, the window $Q = 2N(\sum_i v_i + T_0)$ is met exactly when $\sum_i v_i(c_i - 1) = T_0$ with whole $c_i \geq 1$: unbounded subset sum, which is NP-complete [28], with*

numbers exponential in the size of the instance (the gadgets are of genus one and disconnected). Whether the window alone is strongly NP-hard, with Q polynomial in the size, is open, as is the complexity of maximizing Q over a polytope of lengths in reals.

10 What is machine-checked

The folder `lean` beside this note is a Lean 4 [9] project on Mathlib [32] (both `v4.35.0`). Its 228 theorem declarations, many of them small lemmas, depend only on Lean's standard axioms; the count is not evidence, and its `README.md` maps each statement to its Lean theorems. Lean checks the algebraic and combinatorial core of each statement, with the topological facts it uses (that a loop develops into a given motion, that a curve bounds a disc, that the loops λ_i present the fundamental group, that the quad graph of a sphere is bipartite) as explicit hypotheses. Each statement is formalized on its own abstraction (a turn as a list of motions, a region as finite sets with counts, a system as a pair of matrices): the files share no type of T-mesh and are not composed into one model of it.

Checked, statement by statement. For motions with linear parts in any ring: the Burgers sum, and that the recursion of the gluing maps of a tree of cuts, a hypothesis, unfolds into the order of first visit (Proposition 2(a)); the cut annulus (b); exchanges; the conjugation step of Proposition 8. Over the reals: the four orders and orientations of a dislocation and straight chains (Proposition 3(i), (ii)); staircases; the positivity of a non-negative form with a held arc. At whole lengths: the checkerboard rule; Proposition 5 for whole balanced lengths of any sign, given a 2-colouring at positive lengths and the linearity of α_γ and s_γ; the second proof of Remark 1. The Fox–Burgers formula on the free group with the twisted windings and the insertion of the relation, the case of a simple turn, Proposition 10 and Corollary 2. Lemma 5 on a closed surface; Proposition 17, stronger than stated (on the edges T the whole image of B is the lattice of vectors of even sum); Remarks 2 and 3, with Farkas' lemma over any ordered field and whole multipliers; Theorem 2 and Corollary 3. For fields of any order: the rotation part of Gauss–Bonnet and its counts on triangle and hexagonal meshes, the lengths of dislocations and the congruence of Appendix A.

Checked in a weaker form: Proposition 1, only the additivity of the composed translation, not the integer coefficients nor the reading as a helix; Lemmas 1 and 2, only the Euler-characteristic count on the grid refined by two and the totals from the numbers of convex and concave corners, the regions being discs and the convexity of their corners hypotheses; the completeness of the walks, Proposition 2 of [14], only in a model where the strips are *defined* as the cycles of a permutation of the half-integer stations, so that the bijection holds by construction, with the real walk from $k + \frac{1}{2}$ following the unit edges in a one-dimensional model of the gluing; that the station walk of the gluing data is a permutation, that patch sides are positive and that the model is the quad mesh are assumptions. *Not checked*: the topology taken as hypotheses, strips with borders, the measured statements, and those proved on paper only (see AI assistance). The formalization found no error of sign or convention; it sharpened the hypotheses of Propositions 5, 17 and 6, of Remark 2 and of the completeness proposition, stated in their sharpened form.

11 Open problems and limitations

When is a set of turns closable, in terms of the singularity configuration? Every certificate is a sum of combs, which says nothing about the singularities, and not of staircases, caps and winding separatrices alone; a useful answer lies between, for instance combs on disjoint sets of vertices, each of the sign of the certificate. Computing $\tau_P(H)$ at a given number of quads needs the extremes of an indefinite quadratic function over a polytope. Is Ring-Closing NP-hard for a fixed number of singular vertices? And when the set of turns depends on the lengths, as the strips of the mesh do, what is the problem at all? When does a quartic differential with all

trajectories finite exist for a given singularity configuration [27]? Turns are its discrete version. Finally, a formal model of a complex of rectangles with integer sides, its turns and its developing map would turn the topological hypotheses of the Lean statements into theorems.

Limitations and reproducibility. The measured statements come from one T-mesh of one body, exported by an engine without the constraint rounds of the companion paper, and from a few bench models with an unrecorded engine version; identities were checked in floating point unless exact arithmetic is stated, and the homotopies of Propositions 7 and 8 were not checked. Forced counts are relative to the engine's bounds of separation and, approximately, to its cap, and the intervals depend on solver time limits on a shared machine. The scripts and logs of the exact experiments are not published: the measured statements can be checked against the branch `rd-exact`, an experimental hook of openclare, but not rerun as they stand. The Lean proofs replay with `make proofs NOTE=turns-of-a-t-mesh`. This version of the note was built from commit `6848ef55c9ba`.

AI assistance

This note, its Lean proofs and the software it describes were made by a human author working with an AI system, which is not an author. The author chose the problem and its aims (edge loops that close into rings and, as an artist, few singular vertices and few spirals rather than perfectly square quads), directed the work, decided which results to pursue, and judged them through the measurements and drafts presented to him. Claude Opus 5.5 (Anthropic), used through Claude Code in October 2026 with several sub-agents, wrote the changes to the engine and the experiment scripts, ran the exports and exact computations, searched the literature, stated and proved the propositions, wrote the Lean formalization and drafted the text. Each bibliography entry says how much of the work was read. Beyond the cores machine-checked in Lean (Section 10), everything was written with AI help and checked by AI agents rereading it and testing it on the data, not by a human expert. In particular, Propositions 11, 12, 13, 14, 15 and 16, Lemmas 3 and 4, Theorems 1, 3 and 4, the case of Lemma 5 with borders, and the statements on rotations of order six other than the checked ones have only a paper proof written with AI help. AI review found errors in Propositions 11 and 12 and Theorem 4, and in Proposition 1 and the completeness proposition of [14], whose Lean checks are weaker than the statements; all are corrected here. The note has not been peer reviewed; the author, who publishes it, is responsible for its content.

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A Rotations of other orders

On triangle and hexagonal meshes the rotations are powers of $\zeta = e^{i\pi/3}$, a vertex of valence k has index $6-k$, and a turn along u comes back translated by $T = \alpha u + s\,\zeta u$. The Burgers sum, the order of first visit and the Fox–Burgers formula do not use the order of the rotations and hold unchanged, as do Theorem 2, Corollary 3 and Lemma 5, which use no geometry. A pair of

valences five and seven has $B = \pm\zeta^{\pm1}d$, and the shifts of two turns differ by $\pm d_1$ or $\pm(d_1 + d_2)$, $d = d_1u + d_2\zeta u$. What changes is parity. When every rotation is a power z^{j_i} of one rotation z, every Fox derivative is congruent to its exponent sum modulo $1 - z$, and $1 - z^j$ to $j(1 - z)$ modulo $(1 - z)^2$, so in any commutative ring

$$\tau(w) \equiv (1 - z) \sum_i \epsilon\Big(\frac{\partial w}{\partial x_i}\Big) j_i \, p_i \pmod{(1 - z)^2}.$$

For quarter-turns $(1 - z)^2 = -2z$, and this is Corollary 2. For order six $1 - z$ is a unit: no colouring rule on triangle or hexagonal meshes. When every singular vertex has even valence the rotations have order three, $(1 - \omega)^2 = -3\omega$ (compare Lemma 11 of [19]), and on a closed sphere $\alpha \equiv s \pmod 3$: every ring has a length divisible by three, as on subdivided octahedra, while on subdivided icosahedra rings of $5n$ edges take every residue (measured); a rule modulo three with borders or handles is a conjecture. A dislocation of order n has $|B| = 2\sin(\pi/n)\,|d|$.